
Seminar Task Description: Fish Length Estimation

Deep Learning for Maritime Vision Applications

Topic

This seminar task focuses on estimating fish length from images. The motivation is that, in both commercial fisheries and scientific surveys, manual length measurement is time-consuming and labor-intensive. The project therefore investigates whether deep learning can be used to estimate individual lengths directly from images in a reliable and scalable way.

The main research question is:

How well does an established deep learning baseline perform for fish length estimation, and does replacing its encoder with a Vision Foundation Model (VFM) improve performance, in particular when labeled data are limited?

An optional extension question is:

To what extent does the resulting approach transfer to a second fish dataset?

Expected Scope of the Student Work

The core goal is to compare an established baseline approach with a modified version in which the encoder is replaced by a Vision Foundation Model (VFM).

- defining the prediction target and documenting it precisely,
- preparing the data splits (train/validation/test) and justifying all preprocessing decisions,
- reproducing or approximating the baseline approach as closely as feasible,
- implementing a VFM-based variant with a comparable prediction pipeline,
- evaluating both approaches with suitable regression metrics,
- studying how performance changes when the amount of labeled training data is reduced,
- analyzing failure cases and prediction biases across different samples or species,
- optionally investigating transfer to a second dataset if it becomes available in time.

Provided Data

The main dataset for this seminar task is the *AutoFish* dataset (<https://huggingface.co/datasets/vapaau/autofish>). Use this dataset as the primary basis for model development, comparison, and evaluation. In particular, both the baseline and the VFM-based variant should first be implemented and assessed on AutoFish.

As an optional extension, you may then investigate whether the developed approach also transfers to a second dataset. For this purpose, an additional fish dataset from the North Sea is planned, but it is not yet available at the start of the seminar. If it becomes available in time, it can be used for a small transfer or generalization experiment beyond the main AutoFish-based study.

Starter References

The following references may be useful as starting points for the project:

- S. H. Bengtson, D. Lehotský, V. Ismiroglou, N. Madsen, T. B. Moeslund, and M. Pedersen, "AutoFish: Dataset and Benchmark for Fine-Grained Analysis of Fish," in *2025 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, Tucson, AZ, USA, 2025, pp. 1513–1522, doi: 10.1109/WACVW65960.2025.00174.
- B. Herrmann, E. R. Øye, J. S. Dyrstad, and A. H. Alvestad, "Can automatic measuring replace humans when evaluating a shrimp fishery?," *Regional Studies in Marine Science*, vol. 79, 2024, Art. no. 103852, ISSN 2352-4855, doi: 10.1016/j.rsma.2024.103852.

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